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First-passage time analysis of diffusion-controlled reactions in single-molecule detection

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ABSTRACT

Single-molecule detection (SMD) aims to achieve the ultimate limit-of-detection (LOD) in biosensing. This method detects a countable number of targeted analyte molecules in solution, where the dynamics of molecule diffusion, capturing, identification and delivery greatly impact the SMD's efficiency and accuracy. In this study, we adopt the first-passage time method to investigate the diffusion-controlled reaction process in SMD. We analyze the influence of detection conditions on incubation time and the expected coefficient of variation (CV) under three SMD molecule capturing strategies, including solid-phase capturing (one-dimensional solid-liquid interface fixation), liquid-phase magnetic bead (MB) capturing, and liquid-phase direct fluorescence pair labeling. We find that inside a finite-sized reaction chamber, a finite average reaction time exists in all three capturing strategies, while the liquid-phase strategies are in general more efficient than the solid-phase approaches. CV can be estimated by averaging first-passage time solely in all three strategies, and the CV reduction is achievable given an extended reaction time. To further enable zeptomolar detection, extra treatments, such as adopting liquid-phase fluorescence pairs with high diffusion rates to label the molecule, or designing specific sensing devices with large effective sensing areas would be required. This framework provides solid theoretical support to guide the design of SMD sensing strategies and sensor structures to achieve desired measurement time and CV.

1. Introduction

Single-molecule detection (SMD) holds the promise of achieving the ultimate biosensing detection limit [1]. It improves the limit-of-detection (LOD) by at least 3 orders of magnitude by using the "digital" concept of dividing a sample into small measurable units and suppressing bulk signal noise [2]. Due to this unmatched LOD, SMD is believed to be the next generation of immunoassay technology that could revolutionize fundamental biological studies, clinical diagnosis, and drug developments [3–5]. In recent years, various SMD technologies have been developed, exhibiting sensing capabilities with sub-femtomolar (sub-fM) and even zeptomolar (zM) LOD [6–10].

Despite great progress in the development of sensing devices to resolve single-molecule generated signals, few efforts have been made to investigate the dynamics of molecule capture, identification, and delivery under SMD concentrations [11]. Most SMD technologies rely on Brownian motion for capturing and transporting target molecules, however, the influence of stochastic effects on the total molecule counting efficiency and accuracy is not negligible [1]. When the total number of target molecules present in the solution is under thousands, the questions of "reaction rate barriers" arise on how long the incubation time is required for targeted molecule capturing, which method can secure the delivery efficiency, and, ultimately, whether the counted numbers reflect the solution concentration accurately [2,12,13]. Scattered papers were reported in discussing a few specific SMD sensing structures [14–16], yet, there is a lack of systematic theoretical derivation targeting general particle capturing dynamics at the SMD level.

The first-passage time (FPT) method offers a direct mathematical framework to study the particle diffusion process in SMD [17]. It investigates stochastically the time a single random-walking particle is required to reach a certain target without directly solving cumbersome time-dependent equations [18]. FPT method has been successfully used

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in describing many diffusion-controlled biomolecular processes such as intro/inter-cellular protein transfer, antigen (Ag) - antibody (Ab) reaction, DNA hybridization [19–21], and single molecule tracking in live cells/cell nuclei [22,23]. For instance, single-molecule RelA trajectories were obtained from a microcolony of exponentially growing cells (*E. coli*) to study the intracellular catalytic cycle of RelA [24]. In this case, the sensing channels can be designed in the range of 0.4–0.8 μm height [25] and fabricated for single-cell analysis in microfluidics [26]. Furthermore, it can also be applied to study ecology, such as population migration [27] and habitat selection [28], biophysics (e.g., subdiffusion within proteins [29]), and two loci single particle trajectories analysis [30,31].

Single-molecule detection is evidently a diffusion-controlled process, wherein only countable target molecules are present [32,33]. Thus, FPT method is well suited to the SMD study where each target molecule is analyzed separately. Furthermore, the FPT method provides a way to analyze the expected coefficient of variation (CV) of multiple measurements by calculating the second-order momentum of reaction time, which can be used to predict the accuracy requirements of experimental measurements [34].

In this work, we build a framework for studying the dynamics of molecules in SMD processes using the FPT method. We investigate the influence of detection conditions on the necessary incubation time and expected CV under three distinct SMD molecule capturing strategies: solid-phase capturing, magnetic bead (MB) capturing, and direct fluorescence pair labeling. We discover that inside a finite-sized reaction chamber, a finite average reaction time exists in all three capturing strategies, and that CV can be reduced with an extended reaction time. Further, while liquid-phase strategies are in general more efficient than the solid-phase methods, all three molecule-capture and delivery methods under fM level are theoretically allowed under free diffusion. For zM detection (∼ 60 molecules per mL), the results suggest that additional treatments may be required, such as the use of small-sized liquid-phase fluorescence pairs to label the target molecule in a liquid phase, or the application of special sensing reactors to increase capturing region and refrain molecule motion. This framework provides solid theoretical support to guide the design of SMD sensing strategies and sensor structures to achieve desired measurement time and CV.

2. Methods

An Ag-Ab reaction can be simplified as:

$$A + B \underset{k_{r1}}{\overset{k_D}{\rightleftharpoons}} A \bullet B \underset{k_{r2}}{\overset{k_f}{\rightleftharpoons}} [AB], \tag{1}$$

where k_D denotes the kinetic rate that A and B molecules encounter with a distance close enough to allow chemical transformation, k_f denotes the association rate at which the encountered A and B form the final product AB. k_{r1} and k_{r2} are the dissociation rates of these two steps [13]. The reaction in SMD is a diffusion-controlled process where the reaction rate is limited by the time it takes to bring reactants together by diffusion due to the limited Ag molecule number. In a typical SMD using Ag-Ab reaction for specification, the diffusion-related k_D is normally $10^{-6} \sim 10^{-5} m \bullet s^{-1}$ [35], while the reacton-related k_f is $10^3 \sim 10^7 M^{-1} s^{-1}$ [36–38], which is significantly large. So, by considering the whole process while the reaction rate is limited by the smallest reaction rate constant (or longest reaction time), it is suitable to assume Ag and Ab immediately react once they encounter and k_f can be neglected. Other processes that affect the time consumed before the association of Ag and Ab molecules, such as the crossing of free-energy barrier [39] and rotational diffusion [40,41] are short and negligible. The dissociation process is also considerably long compared with the association process, where the typical dissociation rate constant k_{r2} is $10^{-6} \sim 10^{-3} s^{-1}$, meaning once bounded, an Ag-Ab pair will take hours to separate [36–38]. Thus, the reversible binding of AB can be ignored compared with the association, and the Ag-Ab binding can be considered permanent within the observation window [42,43]. After the capturing of Ag molecules, the time-correlated single photon counting (TCSPC) technique [44,45] can be utilized for signal collection and data statistics. Therefore, the reaction in SMD is simplified to be a one-step process, and the reaction rate can be analyzed by studying the diffusion dynamics of reactants in solution.

A free-moving particle in a homogeneous solution can be described by the diffusion equation [46]:

$$\frac{\partial p(\mathbf{r}, t/\mathbf{r_0}, 0)}{\partial t} = D\nabla^2 p(\mathbf{r}, t/\mathbf{r_0}, 0), \tag{2}$$

where $p(\mathbf{r}, t/\mathbf{r_0}, 0)$ is the distribution function describing the conditional probability of finding the particle at position $\mathbf{r}$ at time t, with the initial position of $\mathbf{r_0}$ at $t = 0$, and D is the homogeneous diffusion coefficient. The FPT distribution in heterogeneity-controlled kinetics can be described by position-dependent diffusivity [47]. It can be used to study specific situations in the biological context [48] (e.g., spatial heterogeneity of the protein diffusivity [49]), or anomalous diffusion considering power-law diffusion coefficient $D(x) \propto |x|^{-\theta}$ ($\theta < 0$ for superdiffusive processes and $\theta > 0$ for subdiffusive processes) [50], such as fast electrons in a hot plasma with DC electric field [51], and turbulent two-particle diffusion in configuration space [52]. In this work, the constant diffusion coefficient is adopted to study the Ag-Ab reaction.

The absorbing boundaries are used to model the reaction, where the particle capturing is represented by the absorption at the boundary and annihilates from the solution. Under this condition, the survival rate $S(\mathbf{r_0}, t)$, which represents the probability that the particle initially at $\mathbf{r_0}$, has not been absorbed at time t, equals to the space integral of the probability distribution function of the particle position:

$$S(\mathbf{r_0}, t) = \int p(\mathbf{r}, t/\mathbf{r_0}, 0) d\mathbf{r}. \tag{3}$$

The mean first-passage time (MFPT) of a particle at an initial position $\mathbf{r_0}$ is given by the time integral of the survival rate:

$$\langle \tau(\mathbf{r_0}) \rangle = \int_0^\infty S(\mathbf{r_0}, t) dt. \tag{4}$$

To obtain $\langle \tau(\mathbf{r_0}) \rangle$, we follow the first-passage time adjoint equation instead of directly solving $p(\mathbf{r}, t/\mathbf{r_0}, 0)$ from Eq. (2) to avoid complicated calculations [46,53]:

$$D\nabla^2 \langle \tau(\boldsymbol{r_0}) \rangle = -1. \tag{5}$$

Eq (5) is in a much-simplified form where MFPT can be directly solved under appropriate boundary conditions.

Considering the probability distribution of the initial position of a particle, the average reaction time independent of the initial position can be written as:

$$\langle \tau \rangle = \int \langle \tau(\mathbf{r_0}) \rangle P(\mathbf{r_0}) d\mathbf{r_0}, \tag{6}$$

where $P(\mathbf{r_0})$ is the probability distribution of the particle's initial position. In the following, we solve the MFPT under one and three dimensions with boundary settings following three practical Ag-capturing strategies.

2.1. One-dimensional model

The 1D model follows the solid-phase capturing strategy. As shown in FIG. 1a, Ab molecules are densely immobilized on a solid surface, and Ag molecules in the solution diffuse to the surface and are captured. This interface detection strategy has been applied in numerous label-free refractive index biosensors [54], field-effect-transistor (FET)

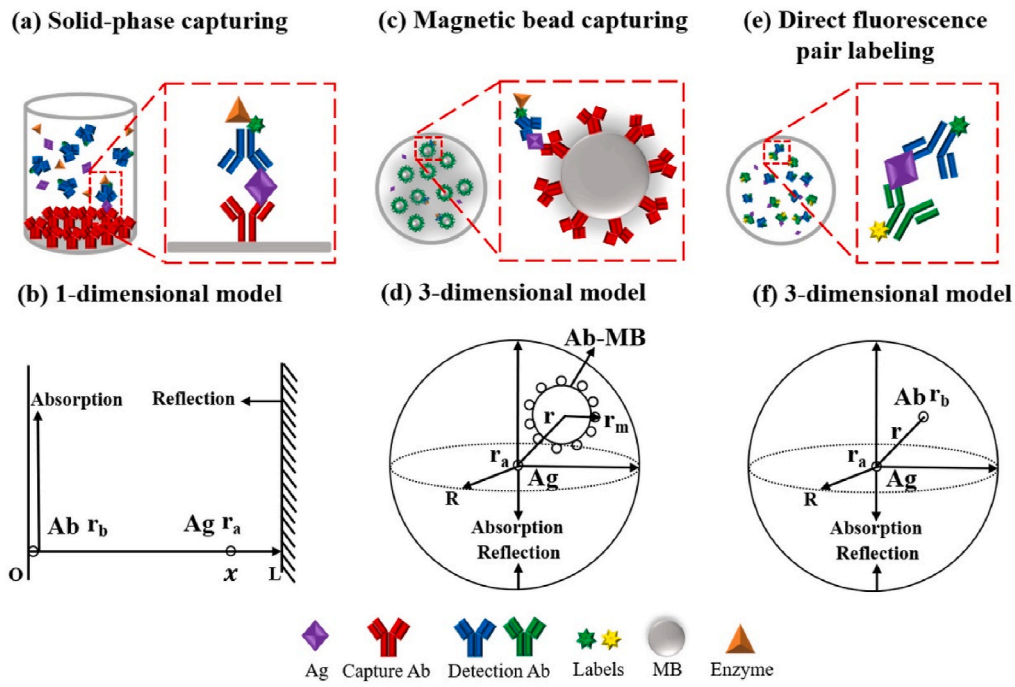

Fig. 1. Schematic illustration of three types of single molecule detection strategies. (a) Solid-phase absorption strategy. The red square is the zoom-in of a capture unit. Free-moving Ag molecules in the solution are captured by Ab molecules densely immobilized on a solid surface. (b) The analysis model of (a). The reactor is a cylinder with an absorption condition at the bottom and a reflection boundary condition at the top and sides. The Ag is a free-moving sphere with a random starting point (uniform distribution). The model is simplified to one dimension. L: liquid column length; r_a: the radius of Ag; r_b: the radius of Ab. (c) The magnetic bead (MB) capturing strategy. Abs are immobilized on MBs to capture Ags. (d) The analysis model of (c). A single spherical Ag with an absorption boundary is located at the center of a spherical reactor with a reflection boundary at its inner surface. An Ab-labeled MB is a free-moving sphere at a random starting point. The MB is partially covered with Abs and there is a ratio that MB will be reflected by the Ag. The model is in three dimensions with spherical symmetry. R the reactor radius; r_m the radius of MB. (e) The direct fluorescence pair labeling strategy. Abs are freely dispersed in the liquid solution to capture the Ags. (f) The analysis model of (e). The setting is similar to (d) but the Ab will not be reflected by the Ag at the center. All mathematical models assume that Ags are immediately captured permanently when they encounter Abs. Considering the typical dimensions of IgG (approximately 14.5 nm × 8.5 nm × 4.0 nm [61]) and protein or viral subunit Ag (< 10 nm [62]), r_a and r_b are set as 5 nm and 2 nm respectively in all calculations if not specifically defined (see Supplementary Information Table S1).

biosensors [55], single molecule electrochemistry detection biosensors [56], and other microfluidic-based biosensors [57], relying on an interface to capture Ag molecules, as well as several commercial SMD instruments developed by Singulex [58,59] and Light and Life. In general, the solid surface area is magnitudes larger than Ag molecules, and there is only one functionalized surface for capturing. Therefore, the reaction can be simplified as a one-dimensional model with the equations to describe the motion of a single Ag molecule [60].

As shown in FIG. 1b, the functional surface is located at zero point whereas the top of the reaction chamber is located at L. An Ab molecule with a radius of r_b is immobilized on the left and one Ag molecule with a radius of r_a walks between a and L, with a random initial location x_0. We apply an absorption boundary at $2a$ and a reflection boundary at L, Eq. (5) can then be transformed to:

$$D\frac{d^2\langle\tau(x_0)\rangle}{dx_0{}^2} = -1 \tag{7}$$

$$\langle\tau(x_0)\rangle|_{x_0=a} = 0$$

$$\frac{d\langle\tau(x_0)\rangle}{dx_0}|_{x_0=L} = 0,$$

where $D = D_a + D_b$, $D_a = k_BT/6\pi\eta r_a$, $D_b = k_BT/6\pi\eta r_b$, k_BT is the product of the Boltzmann constant and the absolute temperature, η is the solvent viscosity, a is the effective radius distance which equals to $(r_a + r_b)$.

The general solution of Eq. (7) is

$$\langle\tau(x_0)\rangle = -\frac{1}{2D}\left(x_0{}^2 + Ax_0 + B\right) \tag{8}$$

$$\langle\tau(x_0)\rangle|_{x_0=a} = 0$$

$$\frac{d\langle\tau(x_0)\rangle}{dx_0}|_{x_0=L} = 0$$

Constants A and B are obtained by substituting the solution into the boundary conditions, and the analytical expression of an average FPT can be rigorously solved:

$$\langle\tau(x_0)\rangle = \frac{a^2 - x_0{}^2}{2D} + \frac{L}{D}(x_0 - a). \tag{9}$$

Assuming the possibility of the Ag's initial position is uniformly distributed in the chamber, we have

$$P(x_0) = \frac{1}{L-a}. \tag{10}$$

Substituting Eqs. 9 and 10 into Eq. (6) and taking the integration over the interval between a and L, we get the rigorous solution of the position-independent MFPT:

$$\langle\tau\rangle = \frac{(L-a)^2}{3D} \tag{11}$$

To consider the size of the functional surface area and the Ag molecule concentration, we assume that the Ag molecule reflects at the side walls. Since diffusion is independent between different dimensions, the 1-dimensional model still applies. Assuming the SMD occurs in a reactor of solid surface area S and liquid column length L, the total number of Ags gives $n_a = LSc_a$, where c_a is the Ag concentration. Thus, the position-independent effective MFPT, considering all the molecules in the sample, is:

$$\langle\tau_{eff}\rangle = \frac{(L-a)^2}{3DLSc_a} \tag{12}$$

2.2. Three-dimensional model

We use a three-dimensional model to study the liquid-phase-based capturing strategies. Two strategies are considered: 1) using Ab-coupled magnetic beads as the capturing host which are commonly used in chemiluminescence immunoassay and have been applied in commercial SMD instruments (FIG. 1c) [16]; 2) using small reporter pairs to directly identify Ag molecules, such as FRET [63], Lumit labeling technology by Promega [15], and luminescent oxygen channeling immunoassay [64] (FIG. 1e). The combination of carbocation intermediate with a nucleophile in single molecule nucleophilic-substitution reaction is another type of liquid-phase-based capturing, which is however heavily dependent on heterolysis process [65].

Different from the solid-phase capturing strategy, here, both Ag and Ab molecules are moving freely in the solution. The model is to fix one particle at the zero point of a spherical coordinate, and use the effective diffusion coefficient $D = D_a + D_b$ to simplify the two-object random walk [66]. Since the concentration of Ag in SMD is orders of magnitude smaller than that of Ab, we can fix the Ag molecule at the zero point and analyze the Ab-carrying particle motion. The Ags are extremely rare in the solution and their interaction is negligible, meaning each Ag bound with an Ab does not affect the local Ab concentration around other Ags. This setting will ease the following analysis when multiple Ab-carrying particles are present.

We set the boundary conditions with spherical symmetry, so we only need to solve Eq. (5) in the radial direction. As shown in FIG. 1d and f, we set the absorbing boundary at $r = a$, and outer boundary at $r = R$ as reflection boundary, representing an Ag molecule at the center of a spherical reactor with a radius of R. Eq. (5) is transformed to Ref. [46]:

$$D\left[\frac{d^2\langle\tau(r_0)\rangle}{dr_0{}^2} + \frac{2}{r_0}\frac{d\langle\tau(r_0)\rangle}{dr_0}\right] = -1 \tag{13}$$

$$\langle\tau(r_0)\rangle|_{r_0=a} = 0$$

$$\frac{d\langle\tau(r_0)\rangle}{dr_0}|_{r_0=R} = 0.$$

Eq. (13) represents a first-order inhomogeneous differential equation for $d\langle\tau(r_0)\rangle/dr_0$, it is easily solved subject to the above boundary conditions. We have:

$$\langle\tau(r_0)\rangle = \int_a^{r_0}\frac{1}{Dx^2}dx\int_x^R y^2dy = \frac{1}{6D}\left(a^2 - r_0{}^2\right) + \frac{R^3}{3D}\left(\frac{1}{a} - \frac{1}{r_0}\right). \tag{14}$$

Assuming there is no interaction between molecules, the positions of all molecules are independently and randomly distributed in a homogeneous spherical reactor, finding the radial distribution function $P(r_0)$ satisfying:

$$P(r_0) = \frac{3r_0{}^2}{R^3 - a^3}. \tag{15}$$

The MFPT equals the integration of the product of $\langle\tau(r_0)\rangle$ and $P(r_0)$ between the radial interval of a and R:

$$\langle\tau\rangle = \int_a^R \tau(r_0)P(r_0)dr_0 = \frac{(R-a)^2}{15Da}\times\frac{5R^3 + 6R^2a + 3Ra^2 + a^3}{R^2 + Ra + a^2}. \tag{16}$$

This analytical formula for average passage times is consistent with previous work, verifying our FPT method for MFPT derivation [46]. We also tested the average nearest-neighbor distances (ANND) to replace r_0 and found the similar results (see Supplementary Information S1) [67].

The 3D model of MB capturing (FIG. 1d) reveals that a single MB has a certain average number (denoted as N) of Ab as binding sites fully or partially covering the bead surface. Instead of treating the absorption and reflection on the MB surface, we add a correction factor to compensate for the reflection. For an MB of radius r_m with each binding site of radius r_s, the fraction of its surface occupied by binding sites is $Nr_s{}^2/4r_m{}^2$. A correction factor $(\pi r_m + Nr_s)\ /Nr_s$ can be added to MFPT derived elsewhere [68]. The effective MFPT considering the MB capturing probability and concentration is

$$\langle\tau_{eff}\rangle = \frac{(R-a)^2}{15Da}\times\frac{5R^3 + 6R^2a + 3Ra^2 + a^3}{R^2 + Ra + a^2}\times\frac{\pi r_m + Nr_s}{Nr_s}\times\frac{1}{Vc_m}, \tag{17}$$

where $V = 4\pi R^3/3$ is the reactor volume, $a = r_m + r_a$, and c_m is the MBs concentration.

The direct reporter pair labeling strategy uses a similar model with an Ag molecule at zero point, but without the correction factor (FIG. 1f). The analytical expression of effective MFPT is

$$\langle\tau_{eff}\rangle = \frac{(R-a)^2}{15Da}\times\frac{5R^3 + 6R^2a + 3Ra^2 + a^3}{R^2 + Ra + a^2}\times\frac{1}{Vc_b}, \tag{18}$$

where c_b is the Abs concentration, the total number of all the Abs is then Vc_b. The reaction rate can be calculated as $f = k_Dc_ac_b$ from the perspective of either macroscopic or microscopic scale [69]. When R approaches infinity, $R \gg a$, the effective MFPT can be simplified to the classic result $\langle\tau_{eff}\rangle \approx 1/4\pi Da = 1/k_D$ [13,70] (see Supplementary Information S1).

To simplify the simulation, we assume the Ab pair molecules have the same size, and their binding sites are far apart without competition, so the two identical capturing processes are separated. The total effective MFPT $\langle\tau_{eff}\rangle$ is simply the sum of two $\langle\tau_{eff}\rangle$ [70], which is also verified by the self-consistent pathway solution considering a generic scheme for single molecule kinetics [71] or serial chemical reactions [72] (see Supplementary Information).

2.3. Expected coefficient of variation

The coefficient of variation (CV) is a crucial parameter to evaluate the accuracy of the results and, more importantly, the reliability of the measurements [73]. Despite instrumental errors, intrinsic measurement uncertainty is governed by the statistical nature of the particle's random walk which is profoundly amplified in SMD where only countable target molecules are present.

Considering the counted molecule number as the measured value in SMD, we analyze two sources of intrinsic uncertainty: 1) the variation of the target molecule number in the testing volume, and 2) the uncertainty of a single particle being captured.

The CV of the molecule number in a certain volume with a given concentration follows the well-known Poisson distribution:

$$CV_N = \frac{\sqrt{N}}{N} \tag{19}$$

where $N = V \times c$ is the expected molecule number, V denotes the sample volume, c denotes the concentration of Ag (c_a) in 1D/3D model.

The probability of a single particle being captured at time t is directly related to the survival rate as $1 - S(t)$. The CV of the particle being captured at t, based on the definition of CV, is (see Supplementary Information S2 for derivation)

$$CV_t = \sqrt{\frac{S(t)}{1 - S(t)}}. \tag{20}$$

Although this term could be solved from Eqs. (2) and (3), to avoid complicity, we took an asymptotic exponential decay to simplify the calculation. For both one-dimensional and three-dimensional models, assuming the particle will eventually be captured (under finite boundary conditions), the survival rate $S(t)$ meets the conditions $S(t)|_{t=0} = 1$ and $S(t)|_{t\to\infty} = 0$, an approximation can be written as [46]

$$S(t) \approx S_{approx}(t) = e^{-\frac{t}{\tau_0}}, \tag{21}$$

where τ_0 is the characteristic decay time which is related to the survival

possibility within the absorbing interval [53]. The discrepancy between this approximation and the rigorous solution is plotted in Fig. S2, showing the identical decay trend (see Supplementary Information S3). By applying Eq. (21), we greatly simplify the expression of the time variation and use one parameter, τ_0, to describe the probability of the molecule hitting the absorption boundary.

To obtain τ_0, we extend the FPT method by using the second-order momentum of the MFPT $\langle\tau^2\rangle$ to get the CV of MFPT from the following relation (see Supplementary Information S4):

$$\sigma(\tau) = \sqrt{\langle\tau^2\rangle - \langle\tau\rangle^2}, \tag{22}$$

$$CV_{\langle\tau\rangle} = \frac{\sigma(\tau)}{\langle\tau\rangle}. \tag{23}$$

Then we match the $CV_{\langle\tau\rangle}$ to Eq (21) to numerically link τ_0 with the MFPT $\langle\tau\rangle$ (Eq. (24))

$$\tau_0 = \frac{\langle\tau\rangle}{\ln\frac{CV_{\langle\tau\rangle}{}^2 + 1}{CV_{\langle\tau\rangle}{}^2}}, \tag{24}$$

and further, calculate the CV under different incubation times.

For the one-dimensional model, we have the rigorous solution of $\langle\tau^2\rangle$ (see Supplementary Information S5). By substituting the solution to Eq. (20), we get $CV_{\langle\tau\rangle} = 1.1$, yielding $S(\langle\tau\rangle) = 0.5455$ and $\langle\tau\rangle = 0.6061\ \tau_0$. For the three-dimensional models, the standard deviation $\sigma(\langle\tau\rangle)$ of the first-passage time is reported to be the same as MFPT $\langle\tau\rangle$ [74]. We also confirmed this result using the discrete FPT method with reasonable accuracy (see Supplementary Information S6). This number directly gives the result of $S(\langle\tau\rangle) = 0.5$, yielding $\langle\tau\rangle = 0.6931\tau_0$.

The total CV is the combination of the two CVs from particle total number and particle captured probability [73]. Since the effective MFPT is simply the production of MFPT and the reciprocal of the particle number in Eqs. (12), (17) and (18), the total CV given the incubation time T, based on the principle of error propagation, can be written as:

$$CV = \sqrt{(CV_T)^2 + (CV_N)^2}. \tag{25}$$

3. Results and discussions

3.1. One-dimensional model: solid-phase capturing strategy

The simulated MFPT in the one-dimensional model is shown in FIG. 2. FIG. 2a shows the MFPT as a function of Ag concentration from zM to pM with various liquid column lengths. The MFPT is inversely proportional to the analyte concentration as expected and increases with the increment of the liquid column length, showing that limited analyte free-moving length accelerates the Ag-Ab reaction rate, because the Ags have a higher chance to move to the solid surface and then be captured by Abs [75]. For detection in extremely low concentrations, e.g., tens of zM [14,76], it may require tens of hours to incubate the mixture of sample and detection reagents before the collection of signals for analog-to-digital analyses.

The influence of sample volume on MFPT is further demonstrated in FIG. 2b. At a certain sample volume, increasing the surface area while decreasing the liquid column length significantly reduces the MFPT for a given Ag concentration, showing a strong preference for a large effective detection area and small height of the reaction chamber (i.e., large S and small L). This structural effect is attributable to the increased binding chance with both large sensing areas and restricted Ag diffusion. Such reactor structures are practically achievable in microfluidic devices where the sensing areas are in micrometer-deep chambers formed by microfabrication technologies.

3.2. Three-dimensional model: MB capturing strategy

The results of the simulated MFPT of MBs capturing strategy are shown in FIG. 3. As illustrated in FIG. 3a, with a fixed MB size, MFPT is inversely proportional to the increment of both the MBs concentration and the average number of Abs on MBs, showing a high concentration of MBs and denser bound Abs lead to a faster reaction. This result is expected since more MBs and denser bound Abs lead to a higher chance of capturing the Ag molecules in solution [77].

The influence of MB size on the effective MFPT is illustrated in FIG. 3b. Under a fixed MB concentration and a given number of capturing sites per MB, the MB size has a small effect on the MFPT, suggesting that the diffusion reduction of large MBs does not significantly affect the capturing chance since it is compensated by the larger capturing size of MBs. The number of the average capturing sites per MB N has a larger influence on MFPT where more capturing sites means a higher capturing chance. N is ultimately limited by the MB surface area. With the same total weight of MBs, smaller MBs are shown to possess a greater total surface area, thus benefiting the capturing process. However, it would also bring magnetic force reduction which negatively affects the separation and wash [78,79]. A trade-off of MB size would need to be considered in experimental practice.

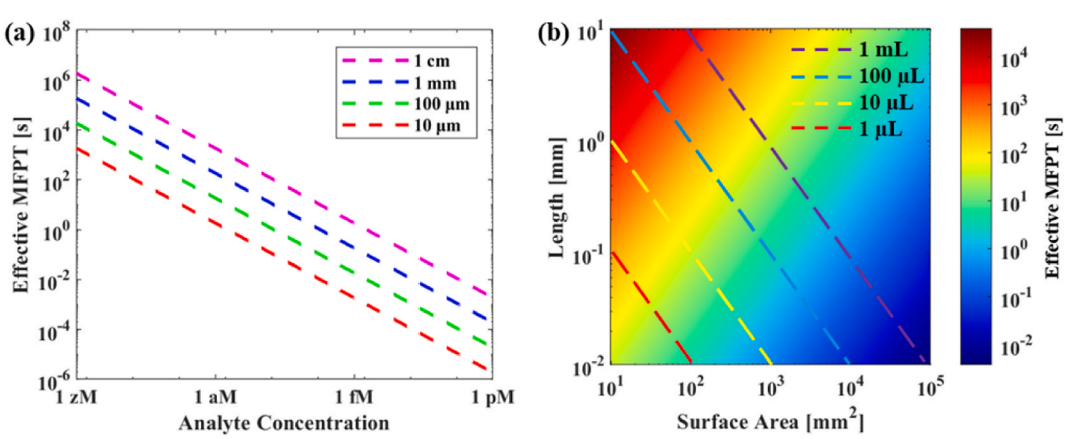

Fig. 2. Simulated results of the solid-phase capturing strategy. (a) The mean first-passage time as a function of analyte concentration with different liquid column lengths. Surface area S is set to a fixed value of 20 mm^2. The MFPT is inversely proportional to the analyte concentration c_a and increases with the increment of the liquid column length L, indicating an accelerated Ag-Ab reaction rate under a limited analyte free-moving length. (b) The map of the MFPT with varied S and L. The MFPT becomes shortened as the surface area is enlarged, indicating a preference for a large effective detection area (i.e., large S and small L). The dotted lines show the same sample volume. c_a: 100 zM.

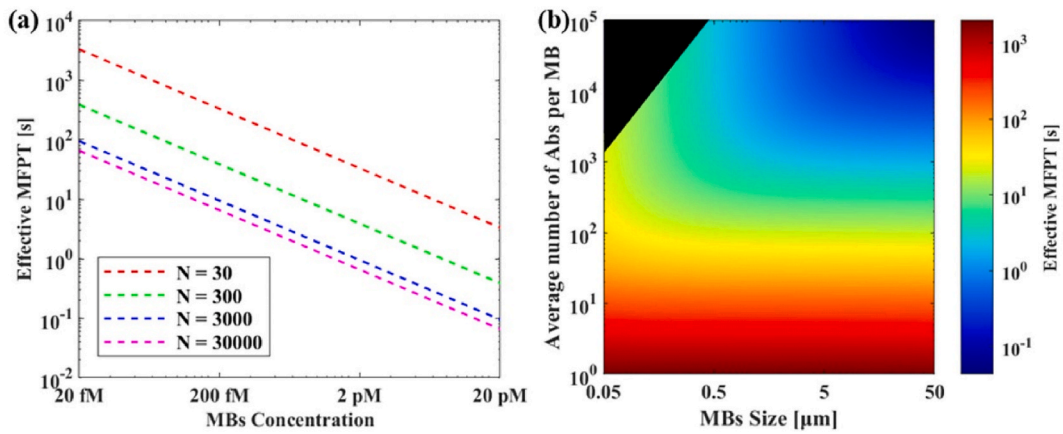

Fig. 3. Simulated results of MB capturing strategy. (a) The MFPT as a function of MBs concentration. N is the average Ab number on each MB. MFPT is inversely proportional to the increment of both the MBs concentration and the average number of Abs on MBs, showing a high concentration of MBs and denser bound Abs lead to a faster reaction. r_m: 1 μm; R: 3 mm. (b) Map of MFPT with varied MB radius r_m and N. Both the MB size and average Ab number on each MB positively increase the reaction rate. The diffusion reduction with a large MB size does not affect the reaction. c_m: 1 pM; R: 3 mm. The black-colored area indicates the upper limit of N.

3.3. Three-dimensional model: direct fluorescence pair labeling strategy

The results of simulated MFPT of direct fluorescence pair strategy are shown in FIG. 4. FIG. 4a shows the same trend in FIG. 3a that the MFPT declines as the Abs concentration is gradually elevated, indicating that a higher Abs concentration leads to a faster reaction. The effect of Ag molecule size could be observed since the size of Ag and Ab molecules is comparable, with the maximum MFPT being obtained when the size of Ag is equivalent to that of Ab. When the size of Ag exceeds 5 nm, a larger capturing target increases the capturing chances as expected. However, this large size of Ag is unfavorable for FRET and Lumit labeling technology since it leads to Ab pair distances beyond 10 nm resulting in a very weak signal or no signal at all [80], yet is acceptable for luminescent oxygen channeling immunoassay [81]. Notably, compared with the MB capturing strategy, although two bindings on one Ag are needed to complete the process, the experimentally applicable concentration of Abs can be orders-of-magnitude larger, which could greatly reduce the MFPT.

The reactor size, as illustrated in FIG. 4b, has a positive effect on reducing the MFPT only when it is smaller than approximately 100 nm. Abs concentration has a more pronounced influence on MFPT, as the high Abs concentration greatly reduces the nearest distance between Ab and Ag molecules. The effect of diffusion limit is observable only when the reactor size is comparable to the molecule size. The corresponding practical reactors could be porous materials such as porous glass and porous silicon.

3.4. Simulated coefficient of variation in SMD

The relationship between MFPT, survival rate, and intrinsic CV_T based on the exponential-decay approximation is shown in FIG. 5a. We use the MFPT $\langle\tau\rangle$ as the unit in x axis to cover various conditions. The corresponding CV_T s in one- and three-dimensional models are slightly different since their $CV_{\langle\tau\rangle}$ s have a 9.5% difference. It is clear that directly using MFPT as the incubation time only provides a 50% chance of capturing the target molecule, yielding a CV_T of 100%, which is consistent with previous work [74]. To achieve a reasonable CV_T, e.g., less than 4% (i.e., $S_T < 0.0016$), the incubation time T should be at least 11 $\langle\tau\rangle$ to guarantee the efficient capturing.

The total CV in the 3D models considering the influence of particle number uncertainty (Eq. (24)) is presented in FIG. 5b, where the particle number is calculated from the sample volume and analyte concentration (the plot with the 1D model is slightly different, see Supplementary Information S5 for calculation). At low analyte concentration (zM ~ aM), the CV is dominated by the uncertainty of the absolute analyte numbers, which is also known as Poisson noise. At relatively high

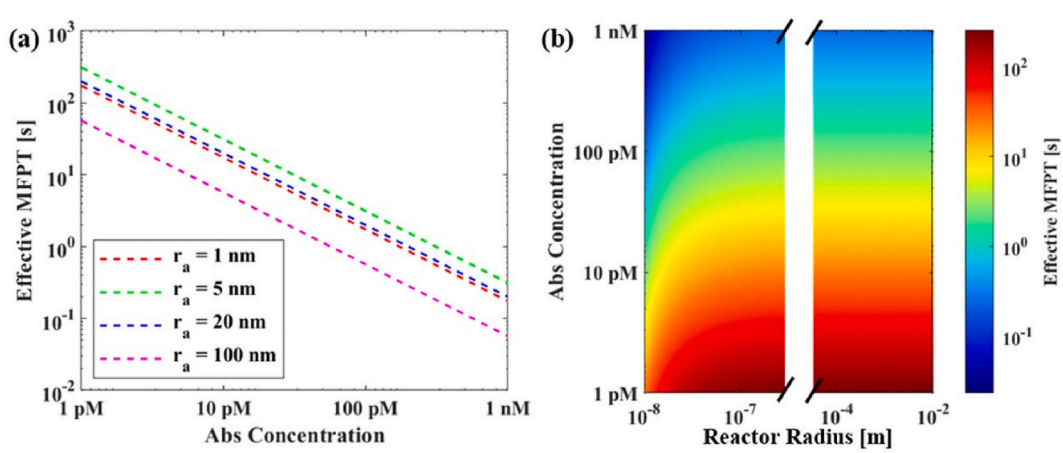

Fig. 4. Simulated results of direct fluorescence pair capturing strategy. (a) The MFPT as a function of Abs concentration (c_b) with a wide range of r_a. The MFPT decreases with the increment of both the Abs concentration and the radius of Ag, indicating that higher Abs concentration and larger Ag molecules lead to a faster reaction. R: 3 mm. (b) Map of MFPT with varied reactor radius R and c_b. The influence of R quickly becomes saturated within tens of nanometers, then the MFPT value depends solely on the Abs concentration.

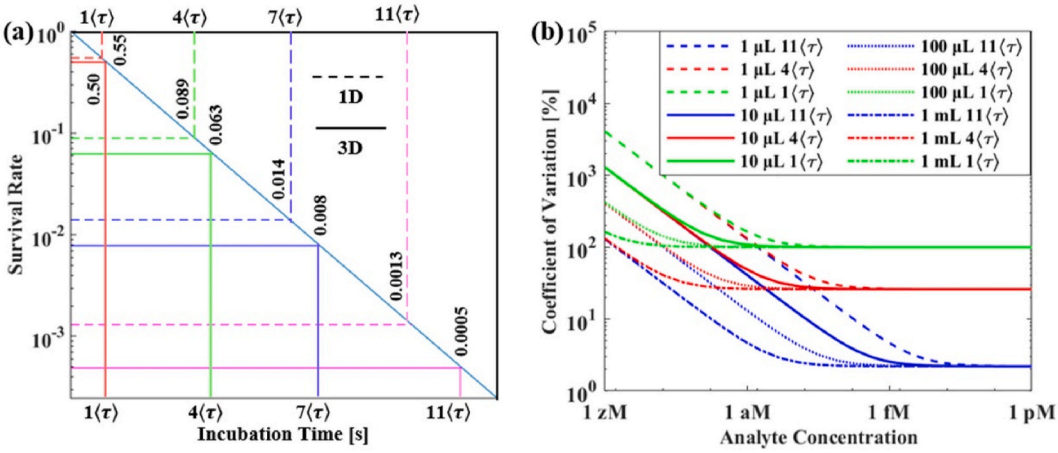

Fig. 5. CV analysis of Ags captured by Abs under all three types of single molecule detection strategies. (a) Analyte molecule survival rate (the missing rate in capturing) in 1D and 3D models as a function of incubation time. ⟨τ⟩ denotes the MFPT. The corresponding survival rates under different ⟨τ⟩ are labeled. The exponential decay of the survival rate as a function of incubation time assumes that the MFPT of molecules diffused in a *d*-dimensional domain with absorption on the boundary follows the Poisson equation. (b) Coefficient of variation vs. analyte concentration (c_a) curves with the reactor radius and incubation time in the 3D models. Under low analyte concentration (zM ~ aM), the CV of MFPT is dominated by the Poisson noise from low absolute analyte numbers (reflected by the sample volume). With relatively high analyte concentration (fM ~ pM), the CV is governed by the incubation time instead of concentration, since the CV is dominated by the average survival rate of each molecule.

analyte concentration (fM ~ pM), the CV is governed by the incubation time where the capturing process plays a more important role. Thus, to suppress the total CV, it is necessary to have a sufficient incubation time in addition to providing sufficient analyte molecule numbers, either by increasing the sample volume or analyte concentration.

Table 1 compares the effective MFPT and corresponding CV simulated using the experimental conditions described in the literature. Here, we use a suggested incubation time of 11 ⟨τ⟩. The calculations assume that initially all the molecules are uniformly distributed in the reactor without stirring. Most of the methods with the LOD ranging from tens of aM to pM show reasonable suggested incubation time and small intrinsic CV, suggesting that statistical uncertainty is not the limiting factor and systematic error from instrumentation is likely to be dominant. When the LOD is reported to be in the scale of zM, the suggested incubation time significantly increases and the intrinsic CV rises to over 40%, suggesting that although the SMD detector is sensitive enough to generate a detectable signal from single analyte molecules, the accountability of the readout needs further investigation by including statistical effects in the reaction.

3.5. Strategies toward zM-scale single molecule detection

To reliably detect molecules on zM scale, both the incubation time and the total analyte molecule number have to be sufficient. Our studies reveal that the liquid-phase (3D) capturing strategies have shorter MFPTs than those of the interface-capturing strategy. For instance, in a simulation with a sample of one Ag in 100 μl solution (~ 10 zM) and an Ab concentration of 100 pM, the MFPT was 25.9 h and 2.5 s for solid-phase capturing (1D) and Ab pair (3D) strategies, respectively. This is due to the fact that, compared with the solid-phase capturing strategy, where the countable Ag molecules have to diffuse to the sensing area, we can add extremely concentrated Ab pairs or MBs in a liquid medium, allowing them to diffuse to the target while reducing the MFPT.

The Ab pair strategy exhibits superior performance compared to the

Table 1
Simulated MFPT and CV of three single molecule detection types[a].

Detection type	Analyte	LOD	Volume (μl)	Incubation time (min)	Simulated CV with incubation time in Ref (%)	Suggested incubation time 11 ⟨τ⟩	Simulated CV with 11 ⟨τ⟩ (%)	Ref
Solid-phase capturing	SARS-CoV-2 RNA	30 zM	100	<4	>1	94.7 h	74.5	[76]
	IgG Ag	90 zM	100	<10	>1	31.5 h	43.1	[14]
	PSA	0.80 fM	100	60	0.5	4.7 s	3.6	[82]
	HIV (VRC01)	5.6 fM	100	120	0.2	0.7 s	3.6	[59]
	SARS-CoV-2 RNA	1.2 aM	20	5	26.3	75.6 s	26.6	[83]
MB capturing	PSA	0.33 pM	100	30	0.01	75.6 s	2.2	[84]
	Tumor Necrosis Factor-a	60 aM	5	1	0.01	0.3 s	7.8	[85]
	LINE1/ORF1 protein	20 aM	400	120	0.006	133.1 s	2.6	[16]
Direct fluorescence pair labeling	Prion protein (PrP)	2.0 nM	100	30	0.0004	2.8 s	2.2	[86]
	Protein G	50 fM	100	30	0.002	55.8 s	2.2	[87]
	IL-6	84 nM	50	90	0.003	55.8 s	2.2	[15]

[a] The values of LOD, volume, and incubation time are obtained from published references; simulated CV with incubation time in Ref, suggested incubation time, and simulated CV with suggested incubation time is calculated from simulation. The calculations assume that initially all the molecules are uniformly distributed in the reactor without stirring.

MB strategy when 3D models are employed and capturing particle numbers are similar. For instance, when their equivalent binding capabilities are assumed to be equal, it was found that 1 pM MBs and 300 average Abs per bead yielded a simulated MFPT of 7.7 s, while 300 pM Ab pairs produced a simulated MFPT of 0.8 s. This difference may be attributed to the angular blocking of Abs on MB. In practice, however, the nonspecific binding of highly concentrated Abs should be taken into account, even though a high concentration of capturing particles is typically desired.

The minimization of the MFPT to enable zM detection is a key challenge. To this end, specifically designed reactors are a promising direction. The 1D model requires a large sensing area with a small reactor height, while the 3D models necessitate even smaller reactor volumes. Micro/nanofluidic structures, such as flat sensing chambers, nanofluidic channels and nanodroplets, may provide suitable platforms for this purpose. However, throughput may become a limiting factor in practical applications.

The detection of zM-scale single-molecule signals requires a sufficient number of analyte molecules to overcome Poisson noise, a physical limitation facing all SMD strategies. To achieve a 10% CV, at least 100 molecules need to be presented, equivalent to a sample of approximately 1.7 mL with 100 zM concentration assuming all molecules are captured. Such a large sample volume requirement impedes the necessary sample separation in SMD detection. To reduce sample size, pre-concentration techniques, such as electrophoretic methods [88], solid-phase extraction [89], and liquid-liquid extraction [90], can be employed. An alternative solution is to use high-throughput fluidic measurements, which can separate and measure a large sample volume, e.g., 2 mL, in a reasonable time [91–93].

4. Conclusion

In this work, we analyze the diffusion-controlled molecule dynamics in SMD processes using the FPT method. Three distinct detection strategies, namely solid-phase capturing, MB capturing, and direct fluorescence pair labeling, are modeled separately by deriving the analytical expression of effective MFPT and CV. Parameters such as Ags/Abs concentration, reactor geometry, footprint, and binding capacity of MBs are studied to assess their influence on SMD performance. We find that a finite MFPT is observed in all three SMD categories within a limited sample amount, and CV can be reduced after a prolonged incubation time. Furthermore, molecule diffusion enables all capturing and delivery below fM detection to be statistically allowed, with liquid-phase strategies generally performing more efficiently than solid-phase approaches. For zM detection, the analysis suggests using liquid-phase fluorescence pairs with a high diffusion rate to label the molecule or designing specific sensing devices with a wide sensing area and thin volume height to accelerate molecule capturing. This framework provides solid theoretical support to guide the design of SMD sensing strategies and sensor structures with the desired measurement time and CV.

Credit author statement

Yingkai Lyu: Writing – original draft, Methodology, Software. **Lixiang An**: Visualization. **Huiyang Zeng**: Validation. **Feng Zheng**: Writing - Reviewing and Editing. **Jiajia Guo**: Writing - Reviewing and Editing. **Pengcheng Zhang**: Writing - Reviewing and Editing. **Hui Yang**: Writing - Reviewing and Editing. **Hao Li**: Supervision, Methodology, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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